SOFTWARE ENGINEERING LAB REPORT

Sequence Diagram

LAB 6 | Case Study: Food Delivery Application

Project Overview

The Food Delivery Application allows customers to discover restaurants, browse menus, place and pay for orders, track delivery, and provide feedback. Restaurants manage menus and incoming orders, while delivery partners accept assignments and update delivery status. An administrator supervises users, restaurants, payments, and operational issues.

Submitted by

Team Member	Roll Number
Deepak Singh	202401100200123
Deepak Maurya	202401100200121
Deepak Nimesh	202401100100122
Deependra Ahirwar	202401100200124

Department of Computer Science & Engineering

1. Aim

To model time-ordered interactions between a customer, application interface, backend services, payment provider, restaurant, and delivery partner.

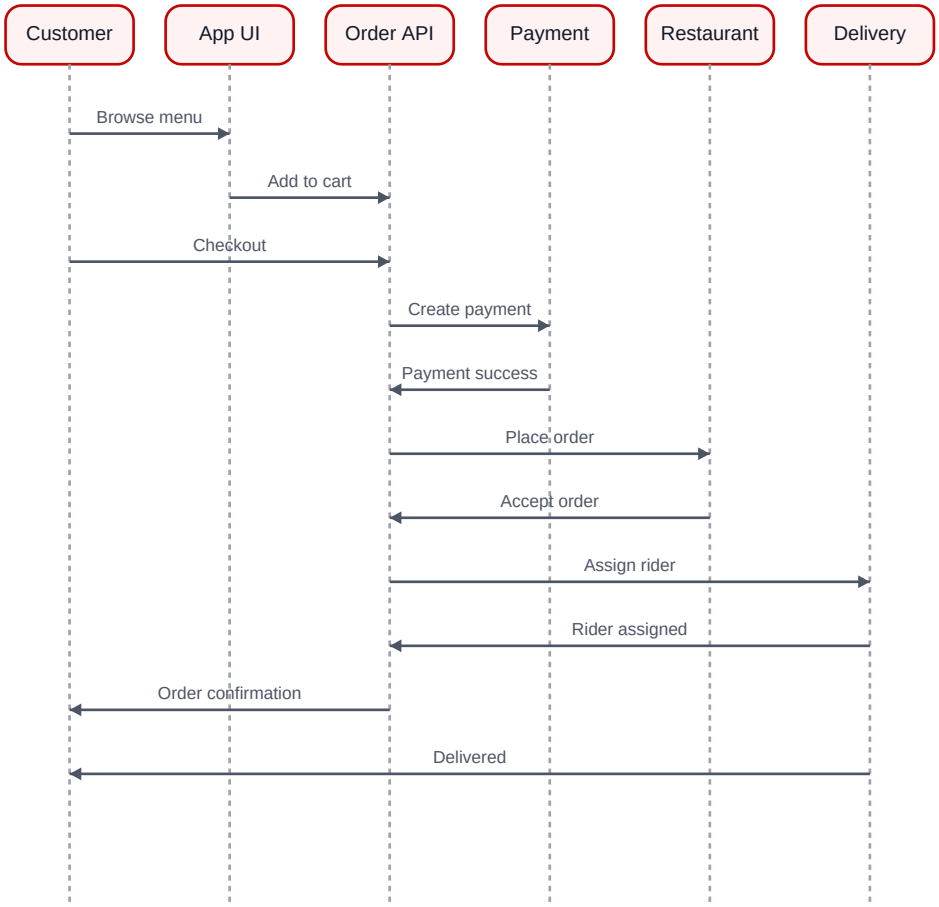
2. Preconditions

• The customer has access to the app and a valid delivery address.

• The restaurant is listed and its menu is available.

• The payment provider and core backend services are reachable.

3. Sequence Diagram



Dashed vertical lines represent lifelines; arrows represent messages.

4. Message Flow

Step	Sender → Receiver	Message / Purpose
1	Customer → App UI	Browse menu and select items
2	App UI → Order API	Submit cart and delivery details
3	Order API → Payment	Create payment request
4	Payment → Order API	Return success/failure and transaction reference
5	Order API → Restaurant	Send order ticket after successful payment
6	Restaurant → Order API	Accept order and update preparation status

Step	Sender → Receiver	Message / Purpose
7	Order API → Delivery	Request rider assignment
8	Delivery → Order API	Return rider assignment and delivery updates
9	Order API → App UI	Publish confirmation, tracking, and final status

5. Result

The sequence diagram demonstrates the main successful checkout interaction. Failure handling should return a clear payment error and must not create a confirmed paid order.